

CS-4820 Database Schemas

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Table 1: accounts

Column	Type	Description
id	UUID (PK)	Unique identifier for all system accounts
account_type	ENUM	User, Researcher, HealthcareProvider, Admin
name	VARCHAR	Account holder name
email	VARCHAR (UNIQUE)	Login identifier
status	ENUM	ACTIVE, DEACTIVATED
created_at	TIMESTAMP	Account creation timestamp
updated_at	TIMESTAMP	Last modification timestamp

Table 2: authentication_credentials

Column	Type	Description
account_id	UUID (PK, FK)	References accounts(id)
password_hash	VARCHAR	Securely hashed password
last_login	TIMESTAMP	Timestamp of last successful login

Table 3: two_factor_methods

Column	Type	Description
id	UUID (PK)	Unique 2FA method identifier
account_id	UUID (FK)	References accounts(id)
method_type	ENUM	SMS, TOTP, Email
secret_key	VARCHAR	Encrypted secret value
enabled	BOOLEAN	Indicates if method is active

Table 4: roles

Column	Type	Description
id	UUID (PK)	Role identifier
role_name	ENUM	User, Researcher, Admin, Provider

Table 5: permissions

Column	Type	Description
id	UUID (PK)	Permission identifier
permission_name	VARCHAR	Named system action
scope	VARCHAR	Data scope for authorization

Table 6: role_permissions

Column	Type	Description
role_id	UUID (FK)	References roles(id)
permission_id	UUID (FK)	References permissions(id)

Table 7: account_roles

Column	Type	Description
account_id	UUID (FK)	References accounts(id)
role_id	UUID (FK)	References roles(id)

Table 8: form_templates

Column	Type	Description
id	UUID (PK)	Template identifier
version	INTEGER	Version number
status	ENUM	DRAFT, PUBLISHED, ARCHIVED
description	TEXT	Template description
created_at	TIMESTAMP	Creation timestamp

Table 9: form_fields

Column	Type	Description
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id	UUID (PK)	Field identifier
form_template_id	UUID (FK)	References form_templates(id)
label	VARCHAR	Display label
field_type	ENUM	Input data type
validation_rules	JSON	Validation constraints

Table 10: form_submissions

Column	Type	Description
id	UUID (PK)	Submission identifier
account_id	UUID (FK)	Submitting user account
form_template_id	UUID (FK)	Template used
status	ENUM	SUBMITTED, FLAGGED, APPROVED
submitted_at	TIMESTAMP	Submission time

Table 11: health_entries

Column	Type	Description
id	UUID (PK)	Health entry identifier
submission_id	UUID (FK)	Originating form submission
account_id	UUID (FK)	Data owner
timestamp	TIMESTAMP	Entry creation time
encrypted_values	JSON	Encrypted health data payload

Table 12: dashboards

Column	Type	Description
id	UUID (PK)	Dashboard identifier
account_id	UUID (FK)	Associated user account

Table 13: health_goals

Column	Type	Description
id	UUID (PK)	Goal identifier
account_id	UUID (FK)	Goal owner
target_value	FLOAT	Target metric value
start_date	DATE	Goal start date
end_date	DATE	Goal end date
status	ENUM	ACTIVE, MET, EXPIRED

Table 14: reports

Column	Type	Description
id	UUID (PK)	Report identifier
researcher_id	UUID (FK)	Creating researcher account
report_type	ENUM	Aggregated, Comparative
created_at	TIMESTAMP	Creation time

Table 15: aggregated_data

Column	Type	Description
id	UUID (PK)	Aggregate record identifier
report_id	UUID (FK)	Parent report
metrics	JSON	Aggregated metrics
anonymization_level	INTEGER	Privacy enforcement level

Table 16: audit_logs

Column	Type	Description
id	BIGINT (PK)	Log entry identifier
actor_id	UUID (FK)	Account performing action
action_type	VARCHAR	Description of action
timestamp	TIMESTAMP	Action time